

ALEXANDROV-TYPE THEOREM FOR SINGULAR CAPILLARY CMC HYPERSURFACES IN THE HALF-SPACE

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(JOINT WORK WITH CHAO XIA)

Alexandrov theorem is a fundamental geometric result, describing the rigidity of closed embedded CMC hypersurfaces. A recent beautiful work [DM19] extends this classical theorem to the sets of finite perimeter setting. In this talk, we consider the rigidity of capillary CMC hypersurfaces in the upper half-space, and extends this result to the sets of finite perimeter setting under certain regularity assumptions.

REFERENCES

- [DM19] Matias Gonzalo Delgadino and Francesco Maggi. “Alexandrov’s theorem revisited”. English. In: *Anal. PDE* 12.6 (2019), pp. 1613–1642. issn: 2157-5045. DOI: [10.2140/apde.2019.12.1613](#). MR: [3921314](#). Zbl: [1418.35205](#).